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## Introduction to Internet of Things

### Assignment- Week 1

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

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#### **QUESTION 1:**

Internet of Things (IoT) can be integrated with which of these separate domains:

- a. Cloud-based storage and computing.
- b. Cyber Physical Systems.
- c. Big-data networks.
- d. All of these.

**Correct Answer: d**

#### **Detailed Solution:**

IoT includes different technologies and all of the mentioned options are essential to consider, which can be integrated with IoT

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#### **QUESTION 2:**

Which of the following is true?

- a. IoT is a subset of M2M
- b. M2M is a subset of IoT
- c. IoT is a subset of CPS
- d. CPS is a subset of Web of Things (WoT)

**Correct Answer: b**

#### **Detailed Solution:**

M2M is a smaller domain and can be consider as part of IoT

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**QUESTION 3:**

M2M is a term introduced by

- a. IoT service provides
- b. Fog computing service providers
- c. Telecommunication service providers
- d. None of these

**Correct Answer: c**

**Detailed Solution:**

See Lecture 1, Slide no 18

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**QUESTION 4:**

The number of possible addresses can be achieved with IPv6 addressing

- a.  $2^6$
- b.  $2^{26}$
- c.  $2^{32}$
- d.  $2^{128}$

**Correct Answer: d**

**Detailed Solution:**

See Lecture 2, Slide 14

---

**QUESTION 5:**

Which of the following is a major concern in IPv4 addressing?

- a. Reliable data transmission



- b. Only addressing
- c. Provide multicast addressing
- d. All of these

**Correct Answer: a**

**Detailed Solution:**

See Lecture 2, Slide 17

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**QUESTION 6:**

Which of these statements regarding sensors is TRUE?

- a. Sensors are input devices.
- b. Sensors can be analog as well as digital
- c. Sensors respond to some external stimuli.
- d. All of these.

**Correct Answer: d**

**Detailed Solution:**

All the options from (a) to (c) are true, thus correct answer will be (d) All of these

---

**QUESTION 7:**

A Transducers

- a. reduce the flow of current in a circuit
- b. convert energy of one kind into another
- c. increase the speed of a motor
- d. convert a sensor to actuator

**Correct Answer: b**



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**Detailed Solution:**

See the lecture on Sensing, slide 4

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**QUESTION 8:**

Potentiometer is which of the following types of sensor?

- a. Speed sensor
- b. Light sensor
- c. Position sensor
- d. Force Sensor

**Correct Answer: c**

**Detailed Solution:**

See the lecture on Sensing, slide 13

---

**QUESTION 9:**

Based on the data type, sensor can be classified in which of the two categories?

- a. Analog and Digital
- b. Isomorphic and Homomorphic
- c. Scaler and Vector
- d. Solid and Liquide

**Correct Answer: c**

**Detailed Solution:**

See the lecture on Sensing, slide 8

---

**QUESTION 10:**



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An actuator is a component of a machine or system, which

- a. senses and transmits the data
- b. moves or controls the mechanism or the system.
- c. only receives the data from other system
- d. None of these

**Correct Answer: b**

**Detailed Solution:**

See the lecture on Actuation, slide 2

---

**QUESTION 11:**

A pneumatic actuator converts

- a. Force in to pressure
- b. Pressure in to acceleration
- c. Pressure in to force
- d. Force in to acceleration

**Correct Answer: c**

**Detailed Solution:**

See the lecture on Actuation, slide 8

---

**QUESTION 12:**

A mechanical actuator converts

- a. rotary motion in to electrical power
- b. electrical power in to rotary motion
- c. rotary motion in to linear motion



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d. Linear motion in to rotary motion

**Correct Answer: c**

**Detailed Solution:**

See the lecture on Actuation, slide 15

---

**QUESTION 13:**

Shape memory polymer (SMP) actuators function similar to

- a. human bone
- b. human muscles
- c. human blood
- d. None of these

**Correct Answer: b**

**Detailed Solution:**

See the lecture on Actuation, slide 18

---

**QUESTION 14:**

Rack and pinion is an example of

- a. Electrical actuator
- b. Hydraulic actuator
- c. Mechanical actuator
- d. None of these

**Correct Answer: c**

**Detailed Solution:**



---

See the lecture on Actuation, slide 15

---

**QUESTION 15:**

Which of the following actuator is used to handle fragile object?

- a. Tiny actuator
- b. Light actuator
- c. Soft actuator
- d. Hydraulic actuator

**Correct Answer: c**

**Detailed Solution:**

See the lecture on Actuation, slide 17

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\*\*\*\*\*END\*\*\*\*\*



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## Introduction to Internet of Things

### Assignment- Week 2

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

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#### **QUESTION 1:**

The MQTT publish-subscribe protocol is:

- a. Event driven
- b. Broker driven
- c. Client driven
- d. All of these

**Correct Answer: a**

**Detailed Solution:**

---

#### **QUESTION 2:**

REST stands for:

- a. Representational State Transfer
- b. Resourceful Sensor Transfer
- c. Resourceful State Transport
- d. None of these

**Correct Answer: a**

**Detailed Solution:**

---

#### **QUESTION 3:**

Brokers in MQTT





- a. Connects publishers and subscribers
- b. Classify the sensor data into topics
- c. Both of the above**
- d. None of the above

**Correct Answer: c**

**Detailed Solution:**

See the lecture on Basics of IoT Networking – Part II, Slide 6

---

**QUESTION 4:**

In AMQP, link initiation in order to send and receive messages is done by which frame type?

- a. OPEN
- b. ATTACH**
- c. BEGIN
- d. FLOW

**Correct Answer: b**

**Detailed Solution:**

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**QUESTION 5:**

MQTT topics are

- a. Simple floating point
- b. Simple integer
- c. Simple symbol
- d. Simple string**

**Correct Answer: d**

**Detailed Solution:**



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See the lecture on Basics of IoT Networking – Part II, Slide 11

---

**QUESTION 6:**

CoAP stands for

- a. Contention Application Protocol
- b. **Constrained Application Protocol**
- c. Cluster Application Protocol
- d. Continuous Application Protocol

**Correct Answer: b**

**Detailed Solution:**

See the lecture on Basics of IoT Networking – Part III, Slide 3

---

**QUESTION 7:**

Which statement is TRUE with respect to the IEEE 802.15.4 standard?

- a. It is a low data-rate standard.
- b. Used for architecting wireless PANs
- c. Uses only two layers – PHY and MAC
- d. **All of these**

**Correct Answer: d**

**Detailed Solution:**

---

**QUESTION 8:**

In the network layer, Zigbee follows which routing scheme?



- a. AODV
- b. DSDV
- c. RIPv2
- d. OSPF

**Correct Answer: a**

**Detailed Solution:**

---

**QUESTION 9:**

XMPP protocol is

- a. Service-oriented middleware based on Hyper Text Markup Language (HTML)
- b. **Message-oriented middleware based on Extensible Markup Language (XML)**
- c. Message-oriented middleware based on Hyper Text Markup Language (HTML)
- d. Service-oriented middleware based on Extensible Markup Language (XML)

**Correct Answer: b**

**Detailed Solution:**

See the lecture on Basics of IoT Networking – Part III, Slide 13

---

**QUESTION 10:**

In AMQP protocol, **Attach** frame type is used to:

- a. Open connection
- b. Informs the changes in state of transfer
- c. **Initiate new link**
- d. Terminate the link

**Correct Answer: c**



**Detailed Solution:**

See the lecture on Basics of IoT Networking – Part IV, Slide 8

---

**QUESTION 11:**

Which of the following is/are the feature(s) of AMQP?

- a. Targeted QoS
- b. Persistence
- c. High speed protocol
- d. **All of these**

**Correct Answer: d**

**Detailed Solution:**

See the lecture on Basics of IoT Networking – Part IV, Slide 11

---

**QUESTION 12:**

IEEE 802.15.4 uses which of the following for accessing the channel?

- a. Carrier sense multiple access (CSMA)
- b. Carrier sense multiple access with collision detection (CSMA-CD)
- c. **Carrier sense multiple access with collision avoidance (CSMA-CA)**
- d. None of these

**Correct Answer: c**

**Detailed Solution:**

See the lecture on Connectivity Technologies – Part I, Slide 6



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**QUESTION 13:**

Which of the following is not true for Reduced Function Device (RFD) in IEEE 802.15.4?

- a. Can only talk to a Full Function Device (FFD)
- b. Can talk to all types of devices**
- c. Lower power consumption
- d. Minimal CPU/RAM required

**Correct Answer: b**

**Detailed Solution:**

See the lecture on Connectivity Technologies – Part I, Slide 10

---

**QUESTION 14:**

Which of the following is true for ZigBee Router (ZR)?

- a. It contains just enough functionality to talk to the parent node, and it cannot relay data from other devices.
- b. This allows the node to be asleep a significant amount of the time thereby enhancing battery life.
- c. Capable of running applications, as well as relaying information between nodes connected to it.**
- d. Memory requirements and cost of ZEDs are quite low, as compared to ZR or ZC.

**Correct Answer: c**

**Detailed Solution:**

See the lecture on Connectivity Technologies – Part I, Slide 22

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**QUESTION 15:**

RPL Routing



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- a. Maintains routing topology using high rate beaconing
- b. Maintains routing topology using low rate beaconing**
- c. Maintains routing topology without beaconing.
- d. None of these

**Correct Answer: b**

**Detailed Solution:**

See the lecture on Connectivity Technologies – Part II, Slide 13

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\*\*\*\*\*END\*\*\*\*\*



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## Introduction to Internet of Things

### Assignment- Week 3

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

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#### **QUESTION 1:**

In Near Field Communication (NFC), the active devices are able to

- a. only collect information
- b. only transmit information
- c. collect and transmit information
- d. only receive information

**Correct Answer: c**

**Detailed Solution:**

See lecture on Connectivity Technologies – Part III, Slide no 15

---

#### **QUESTION 2:**

The Physical layer of HART protocol is derived from

- a. IEEE 802.15.2
- b. IEEE 802.15.4
- c. IEEE 802.15.6
- d. IEEE 802.16

**Correct Answer: b**

**Detailed Solution:**

See lecture on Connectivity Technologies – Part III, Slide no 5

---



**QUESTION 3:**

Near Field Communication (NFC) is based on the principle of

- a. Reflection
- b. Refraction
- c. Magnetic induction
- d. Electrical conductance

**Correct Answer: c**

**Detailed Solution:**

See lecture on Connectivity Technologies – Part III, Slide no 16

---

**QUESTION 4:**

Inquiry run by one Bluetooth device to

- a. Form a connection with another device
- b. discover other devices near it
- c. participate in the network
- d. isolate from the network

**Correct Answer: b**

**Detailed Solution:**

See lecture on Connectivity Technologies – Part IV, Slide no 6

---

**QUESTION 5:**

The sniff mode in Bluetooth is used to

- a. Actively transmitting or receiving data
- b. Sleep and only listens for transmissions at a set interval
- c. Switch in power-saving mode where a device sleeps for a defined period and then returns back to active mode
- d. Deactivate the slave

**Correct Answer: b**





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**Detailed Solution:**

See lecture on Connectivity Technologies – Part IV, Slide no 7

---

**QUESTION 6:**

Which of the following layers provides connection-oriented and connectionless data services to upper layer protocols in blue tooth?

- a. L2CAP
- b. Baseband
- c. Physical radio
- d. Application

**Correct Answer: a**

**Detailed Solution:**

See lecture on Connectivity Technologies – Part IV, Slide no 8

---

**QUESTION 7:**

Which of the following modulations is utilized by Zwave?

- a. Amplitude Phase Shift Keying (APSK)
- b. Quadrature Phase Shift Keying (QPSK)
- c. Frequency Shift Keying (FSK)
- d. Gaussian Frequency Shift Keying (GFSK)

**Correct Answer: d**

**Detailed Solution:**

See lecture on Connectivity Technologies – Part V, Slide no 5

---

**QUESTION 8:**

The PHY and MAC layers in ISA 100.11A are based upon;

- a. IEEE 802.11a



- b. IEEE 802.15.4
- c. IEEE 802.15.6
- d. IEEE 802.11g

**Correct Answer: b**

**Detailed Solution:**

---

**QUESTION 9:**

Event aware topology management in WSNs include:

- a. Timely detection of an event of interest
- b. Monitoring the event
- c. Adapting with the changes of event state
- d. All of these

**Correct Answer: d**

**Detailed Solution:**

---

**QUESTION 10:**

WBAN stands for:

- a. Wireless Buffer Area Networks
- b. Wireless Body Area Networks
- c. Wired Body Area Networks
- d. Wired Buffer Area Networks

**Correct Answer: b**

**Detailed Solution:**

---

**QUESTION 11:**

In a wireless sensor network, a node which are unable to perform an operation, because of power failure and environmental effects is known as

- a. normal node
- b. selfish node



- c. failed node
- d. badly failed node

**Correct Answer: c**

See lecture on sensor networks Part II, slide no 3

**Detailed Solution:**

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**QUESTION 12:**

A dumb node

- a. can sense its surrounding and is able to transmit the sensed data
- b. cannot sense its surrounding, but it can transmit the data
- c. can sense its surrounding, but unable to transmit the sensed data
- d. cannot sensor its surrounding and is unable to transmit the sensed data

**Correct Answer: c**

**Detailed Solution:**

See lecture on sensor networks Part II, slide no 5

---

**QUESTION 13:**

Which of the following is a scheme for re-establishing the connectivity in the presence of dumb nodes in a wireless sensor noetworks?

- a. INTSEM
- b. CoRD
- c. SMAC
- d. DutyCon

**Correct Answer: b**

**Detailed Solution:**

See lecture on sensor networks Part II, slide no 6

---

**QUESTION 14:**



---

The HART protocol was designed for:

- a. Home automation networking
- b. Networked Smart Field devices
- c. Vehicular networking
- d. All of these

**Correct Answer: b**

**Detailed Solution:**

---

**QUESTION 15:**

INTSEM controls the transmission rate of a node by adjusting

- a. a node's data transmission rate
- b. a node's sleep time
- c. a node's receiving rate
- d. the bandwidth of the channel

**Correct Answer: b**

**Detailed Solution:**

See lecture on sensor networks Part II, slide no 8

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\*\*\*\*\*END\*\*\*\*\*



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## Introduction to Internet of Things

### Assignment- Week 4

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

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#### **QUESTION 1:**

UAV-VANETS link-up may be used for:

- a. visual guidance
- b. Data-muling
- c. Coverage enhancement
- d. **All of these**

**Correct Answer: d**

**Detailed Solution:**

---

#### **QUESTION 2:**

In which of the following frequency range does nano networks work?

- a. radio waves
- b. micro waves
- c. **terahertz waves**
- d. infrared

**Correct Answer: c**

**Detailed Solution:**

Nano networks utilizes electromagnetic communication among nano-devices in the terahertz band (0.1-10 THz).

<https://en.wikipedia.org/wiki/Nanonetwork>

---

#### **QUESTION 3:**



In which of the following communication in nano networks, **Gap** junction serves as a mediator between cells and vesicles?

- a. Electromagnetic
- b. Acoustic
- c. **Molecular**
- d. Optical

**Correct Answer: c**

**Detailed Solution:**

See lecture on Sensor Networks Part III, Slide no 11

---

**QUESTION 4:**

Which of the following is true?

- a. In stationary wireless sensor networks, the creation of cluster node is not possible
- b. **In stationary wireless sensor networks, node failure may results in partition of networks**
- c. The topology maintenance is very hard task in a stationary wireless sensor network.
- d. None of the above is true

**Correct Answer: b**

**Detailed Solution:**

See lecture on Sensor Networks Part V, Slide no 2

---

**QUESTION 5:**

In Human-centric sensing, which of these is NOT a constraint?

- a. Energy of sensing devices
- b. Privacy of users
- c. Participant selection
- d. **Network size**

**Correct Answer: d**



---

**Detailed Solution:**

---

**QUESTION 6:**

The main goal of participatory sensing is

- a. just to collect data
- b. to restrict the common people to access data
- c. collect data and allow common people to access data and share knowledge**
- d. to collect data and restrict people to access data and share knowledge

**Correct Answer: c**

**Detailed Solution:**

See lecture on Sensor Networks Part V, Slide no 22

---

**QUESTION 7:**

Which of these topologies cannot be a reliable UAV network topology

- a. Star
- b. Flat mesh
- c. Hierarchical mesh
- d. Tree**

**Correct Answer: d**

**Detailed Solution:**

---

**QUESTION 8:**

Which of the following is true for UAV networks?

- a. links break frequently in it
- b. It is very complex
- c. It needs Huge power requirements
- d. All of the above**

**Correct Answer: d**

**Detailed Solution:**



---

See lecture on Sensor Networks Part V, Slide no 22

---

**QUESTION 9:**

Machine-to-Machine (M2M) is designed for

- a. isolated systems using proprietary solutions
- b. cross platform integration**
- c. home automation only
- d. none of the above

**Correct Answer: b**

**Detailed Solution:**

See lecture on Machine to Machine communication, Slide no. 2

---

**QUESTION 10:**

Which of these can be regarded as the salient feature(s) of a 3D UWSN architecture?

- a. Silent & energy-efficient scheme for mobile UWSNs
- b. Iterative approach with less initiators nodes (anchors) required
- c. Mobility prediction
- d. All of these**

**Correct Answer: d**

**Detailed Solution:**

See lecture-16, slide 15

---

**QUESTION 11:**

Low-end M2M nodes are:

- a. Mobile
- b. Energy efficient**
- c. Complex
- d. Costly





---

**Correct Answer: b**

**Detailed Solution:**

---

**QUESTION 12:**

In M2M, which of the following is true?

- a. Low-end sensor nodes are mobile
- b. Both mid-end and high-end sensor nodes are static
- c. Both low-end and high-end sensor nodes are mobile
- d. High-end sensor nodes are mobile**

**Correct Answer: d**

**Detailed Solution:**

See lecture on Machine to Machine communication, Slide nos. 8-10

---

**QUESTION 13:**

Which of the following is/are the challenge(s) of human-centric sensing?

- a. Energy of devices
- b. Participant selection
- c. Privacy of users
- d. All the above**

**Correct Answer: d**

**Detailed Solution:**

See lecture on Sensor Networks Part V, Slide no 21

---

**QUESTION 14:**

OGDC stands for:

- a. Optimal Geographical Destination Control
- b. Optimal Geographical Density Control**
- c. Optimal Geographical Destination Communication
- d. Optimal Geographical Density Communication



**Correct Answer: b**

**Detailed Solution:**

See lecture on Sensor Networks Part IV, Slide no 15

---

**QUESTION 15:**

In M2M communications, which of the following nodes is/are typically used for general environmental monitoring?

- a. **Low-end sensor nodes**
- b. Mid-end sensor nodes
- c. High-end sensor nodes
- d. All the above

**Correct Answer: a**

**Detailed Solution:**

See lecture on Machine to Machine communication, Slide no. 8

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\*\*\*\*\*END\*\*\*\*\*



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## Introduction to Internet of Things

### Assignment- Week 5

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

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#### **QUESTION 1:**

Why interoperability is required in Internet of Things?

- a. Heterogeneous wireless communication protocols
- b. Different programming languages
- c. Different control models
- d. All of the above

**Correct Answer: d**

**Detailed Solution:** See lec 21, Slide 5

---

#### **QUESTION 2:**

Unique identification codes associated with device interoperability are:

- a. Electronic Product Code (EPC)
- b. Universal Product Code (UPC)
- c. Uniform Resource Identifier (URI)
- d. All of these

**Correct Answer: d**

**Detailed Solution:** see lec 21, slide 11.

---

#### **QUESTION 3:**

The interoperability between devices and device users in terms of syntax is termed as:

- a. Syntactic interoperability
- b. Synthetic interoperability



- c. Both a and b
- d. None of these

**Correct Answer: a**

**Detailed Solution:** see lec 21, slide 13.

---

**QUESTION 4:**

The UMB-A adaptor:

- a. Convert physical devices into virtually abstracted ones
- b. converts local middleware's message into global metadata's message
- c. both a and b
- d. none of these

**Correct Answer: c**

**Detailed Solution:** see lecture 21, slide 21

---

**QUESTION 5:**

Arduino UNO supports output supply voltage of:

- a. 5V
- b. 3.3 V
- c. 9 V
- d. 5V, 3.3V

**Correct Answer: d**

**Detailed Solution:** it is as per Arduino UNO specifications and hardware architecture.

---

**QUESTION 6:**

What is the servo library command to move the shaft of Servo motor back and forth?

- a. Knob()
- b. Sweep()
- c. write()
- d. detach()



**Correct Answer: c**

**Detailed Solution:** write() is the command to tell the servo motor to rotate.

---

**QUESTION 7:**

An Arduino UNO has following number of analog inputs:

- a. 2
- b. 4
- c. 6
- d. 8

**Correct Answer: c**

**Detailed Solution:** it is as per Arduino UNO specifications and hardware architecture.

---

**QUESTION 8:**

What is the functionality of the following code?

```
void setup() {  
    Serial.begin(9600);  
}
```

```
void loop() {  
    int sensorValue = analogRead(A0);  
    Serial.println(sensorValue);  
    delay(1);  
}
```

- a. Reads analog values from A0 only.
- b. Reads analog values from A0 and prints on the serial port/terminal only.
- c. Reads analog values from A0 and prints on the serial port over and over again millisec. after a delay of 1
- d. None of these.

**Correct Answer: c**



---

**Detailed Solution:** Reads analog values from A0 and prints on the serial port over and over again after a delay of 1 millisc.

---

**QUESTION 9:**

What is the functionality of the following code?

```
#include <Servo.h>

Servo myservo;
int potpin = 0; // potentiometer value initialization
int val;

void setup() {
  myservo.attach(9);
}

void loop() {
  val = analogRead(potpin);
  myservo.write(val);
  delay(15);
}
```

- a. Reads potentiometer values on A0 only.
- b. Reads potentiometer values on A0 and writes to servo controller in a loop after a delay of 15 milliseconds.
- c. Writes values to servo motors only.
- d. None of these

**Correct Answer: b**

**Detailed Solution:** The program reads potentiometer values from pin A0 and writes it to the servo, making it rotate. New values are written every 15 milliseconds.

---

**QUESTION 10:**

Pin 4 of a DHT sensor is connected to which pin of Arduino?

- a. 5V
- b. GND



- c. A0
- d. D0

**Correct Answer: b**

**Detailed Solution:** Pin 4 of a DHT sensor should always be connected to the GND pin as per the sensor's design.

---

**QUESTION 11:**

The function random(max) in Arduino gives:

- a. A random number between 1 and max.
- b. A random number between 0 and max.
- c. A random number sequence of length equal to max.
- d. None of these.

**Correct Answer: b**

**Detailed Solution:** See lecture 23, slide 14

---

**QUESTION 12:**

ISR in Arduino stands for:

- a. Interrupt service routine
- b. Information service route
- c. Interrupt service route
- d. Information server route

**Correct Answer: a**

**Detailed Solution:** See lecture 23, slide 15. ISR stands for Interrupt Service Routine.

---

**QUESTION 13:**

**Which of the following is not a valid arithmetic operator in Arduino?**



- 
- a. \*
  - b. x
  - c. +
  - d. -

**Correct Answer: b**

**Detailed Solution:** 'x' is not a valid arithmetic operator as per arduino language.

---

**QUESTION 14:**

The function delay(xx) counts the values of delay in:

- a. Seconds
- b. Minutes
- c. Milliseconds
- d. Hours

**Correct Answer: c**

**Detailed Solution:** The delay function in Arduino programming provides delay in milliseconds

---

**QUESTION 15:**

The number of digital I/O pins in Arduino UNO is:

- a. 6
- b. 4
- c. 8
- d. 14

**Correct Answer: d**

**Detailed Solution:** it is as per Arduino UNO specifications and hardware architecture.





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\*\*\*\*\*END\*\*\*\*\*



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## Introduction to Internet of Things

### Assignment- Week 6

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

---

#### **QUESTION 1:**

d = {1:'item','k':2} data type in python represents:

- a. Dictionary
- b. Array
- c. String
- d. List

**Correct Answer: a**

**Detailed Solution:** See lec 26, Slide 6

---

#### **QUESTION 2:**

The output of the following code snippet is:

```
from math import pi  
print (pi)
```

- a. 3.14
- b. 2.73
- c. pi
- d. None of these

**Correct Answer: a**

**Detailed Solution:** print(pi) prints the value of pi, which is a pre-defined constant.

---

#### **QUESTION 3:**

The following python snippet does the following task:



---

```
file=open('data.txt', 'r')
```

- a. Opens a file (data.txt) in write mode
- b. Opens a file (data.txt) in read mode**
- c. Opens a file (data.txt) in append mode
- d. None of these

**Correct Answer: b**

**Detailed Solution:** 'r' mode is known as the read mode.

---

#### **QUESTION 4:**

The Raspberry Pi3 has how many GPIO pins?

- a. 20
- b. 30
- c. 40**
- d. 50

**Correct Answer: c**

**Detailed Solution:** as per hardware specifications, raspberry pi 3 has 40 GPIO pins. Also refer to <https://www.raspberrypi.org/documentation/usage/gpio/>

---

#### **QUESTION 5:**

The following python snippet does the following task:

```
file=open('data.txt', 'r+')
```

- a. Opens a file (data.txt) in write mode
- b. Opens a file (data.txt) in read mode
- c. Opens a file (data.txt) in append mode
- d. Both a and b**

**Correct Answer: d**

**Detailed Solution:** 'r+' mode is can work as both read/write modes.



---

**QUESTION 6:**

The raspberry pi zero has the following core specification associated with it?

- a. ATMEGA32
- b. PIC16
- c. **ARM-11**
- d. Quad Cortex A53

**Correct Answer: c**

**Detailed Solution:** see lec28, slide3.

---

**QUESTION 7:**

The command *raspistill -o image.jpg* on a Raspberry Pi terminal:

- a. Captures an image
- b. Stores an image from any camera
- c. Reads an image
- d. **Captures and stores an image using the raspberry pi camera**

**Correct Answer: d**

**Detailed Solution:** see lec-29, slide 16.

---

**QUESTION 8:**

What are the types of GPIO pin modes in Raspberry pi 2/3?

- a. Board.
- b. BCM.
- c. **Both a and b.**
- d. None of these

**Correct Answer: c**

**Detailed Solution:** raspberry pi 2 and 3 supports only two pin modes – Board and BCM. The pin configuration changes according to these modes.



---

**QUESTION 9:**

What is the functionality of the following python code?

```
import time  
i = 0;  
def function_a(k):  
    i += k  
    time.sleep(5)  
    print i
```

```
function_a(1)
```

- a. Prints 1 on the console after 5 seconds
- b. Prints 5 on the console after 5 milliseconds
- c. Prints 1 on the console after 5 seconds
- d. Prints 5 on the console after 1 second.

**Correct Answer: a**

**Detailed Solution:** The function call argument is 1 in the last line of the snippet. Upon entering the function, i is added to 1, and as i is 0, the print value is 1. Prior to this a delay of 5 seconds is executed.

---

**QUESTION 10:**

Which one is correct for the nested controlling statement in python?

- a. elseif
- b. elif
- c. ifwhile
- d. none

**Correct Answer: b**

**Detailed Solution:** elif is the valid nested controlling statement in python.

---

**QUESTION 11:**



---

Which of the following is mutable object in python?

- a. Tuple
- b. Bool
- c. Int
- d. List**

**Correct Answer: d**

**Detailed Solution:** mutable object can be changed after it is created, and an immutable object can't. Objects of built-in types like (int, float, bool, str, tuple, unicode) are immutable. Objects of built-in types like (list, set, dict) are mutable

---

**QUESTION 12:**

Internet of Things refers to:

- a. Interconnection of devices in a network
- b. Exchange of data between devices in a network
- c. An ecosystem of devices, accessible through internet
- d. All of the above**

**Correct Answer: d**

**Detailed Solution:** all options a,b,and c satisfy the definition of IoT.

---

**QUESTION 13:**

What is the maximum peripheral current draw allowed in Raspberry pi 3?

- a. 1200mA**
- b. 700mA
- c. 500mA
- d. 100mA



---

**Correct Answer: a**

**Detailed Solution:** As per hardware specifications of raspberry pi.

---

**QUESTION 14:**

The raspberry pi 3 has :

- a. **400MHz video core IV**
- b. 250 MHz video core IV
- c. 100 MHz video core IV
- d. No video core

**Correct Answer: a**

**Detailed Solution:** As per hardware specifications of raspberry pi.

---

**QUESTION 15:**

**A raspberry pi has:**

- a. An inbuilt OS on the board itself.
- b. **An OS on a memory card.**
- c. Both a and b
- d. None of these

**Correct Answer: b**

**Detailed Solution:** the raspberry pi OS is hosted on an external memory card. There is no provision for an inbuilt memory on the raspberry pi as of now.

---

\*\*\*\*\*END\*\*\*\*\*



---

## Introduction to Internet of Things

### Assignment- Week 7

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

---

#### **QUESTION 1:**

Protocol used for connection less socket programming:

- a. TCP
- b. UDP**
- c. Both
- d. None

**Correct Answer: b**

**Detailed Solution:** TCP is known as a connection-oriented protocol, whereas UDP is connectionless.

---

#### **QUESTION 2:**

Which one is an SDN controller:

- a. POX
- b. OpenDayLight
- c. ONOS
- d. All of the above**

**Correct Answer: d**

**Detailed Solution:** see lecture 34, slide 16.

---





**QUESTION 3:**

Give the output of the following snippet:

*Name= "Split-this-string"*

*Name.split("-")*

- a. ["Split", "this", "string"]
- b. Split, this, string
- c. ["Split-", "this-", "string"]
- d. Split this string

**Correct Answer: a**

**Detailed Solution:** The splitting operation is done on the basis of "-". The resultant is optionA.

---

**QUESTION 4:**

Which of the following is not a function in matplotlib library of Python:

- a. display()
- b. show()
- c. subplot()
- d. title()

**Correct Answer: a**

**Detailed Solution:** Show(), Subplot(), Title() are all parts of the matplotlib library. Display ( is not part of this library).

---

**QUESTION 5:**

Application program interface (API) used between application and control planes is known as:

- a. Northbound API
- b. Eastbound API
- c. Southbound API
- d. Westbound API



**Correct Answer: a**

**Detailed Solution:** see lecture 34, slide 3

---

**QUESTION 6:**

The bind() function in Python socket programming is used to:

- a. **Specify the port for service on the specified host**
- b. Read data from the socket
- c. Send data to the socket
- d. Initiate the connection

**Correct Answer: a**

**Detailed Solution:** see lecture 31, slide 13.

---

**QUESTION 7:**

Opening a file in append mode in Python:

- a. Overrides the existing data in the file
- b. **Adds the new data to the end of the existing file**
- c. Adds the new data at the beginning of the existing file
- d. Cannot write data to a file in append mode

**Correct Answer: b**

**Detailed Solution:** append means adding to an existing file without deleting the original contents. This is what the append mode in Python does too.

---

**QUESTION 8:**

A typical socket program on raspberry pi using python:

- a. Creates a two-way communication between two nodes in a network
- b. Server performs the task/service requested by the client
- c. **Both a and b**
- d. None of these

**Correct Answer: c**



---

**Detailed Solution:** see lec-31, slide 11.

---

**QUESTION 9:**

In Python based socket programming, which of these is/are represent a socket family?

- a. AF\_UNIX
- b. AF\_INET
- c. **Both a and b**
- d. None of these

**Correct Answer: c**

**Detailed Solution:** See lec31, slide 12.

---

**QUESTION 10:**

In Python based socket programming, which of these is/are represent a socket type?

- a. AF\_UNIX
- b. AF\_INET
- c. **SOCK\_STREAM**
- d. SOCK\_DATA

**Correct Answer: c**

**Detailed Solution:** See lec31, slide 12.

---

**QUESTION 11:**

Arrange the following events sequentially (starting from the first step to the last) for sending data from client to server using sockets and saving it?

- A. *Create a client and server*
- B. *Establish connection between the two*
- C. *Send data from the client to the server*
- D. *Save the data in a file*



- a. A-B-C-D
- b. A-C-D-B
- c. C-B-D-A
- d. None of the sequences are true

**Correct Answer: a**

**Detailed Solution:** see lec-32, slide 9

---

**QUESTION 12:**

The `ion()` function in matplotlib plotting:

- a. Adds lines to the plot
- b. makes subplots in the plot area
- c. provides title to the plot
- d. turns on interactive plot mode**

**Correct Answer: d**

**Detailed Solution:** see lec-32, slide 15.

---

**QUESTION 13:**

The following are the limitations of our current traditional networks:

- a. Vendor-specific architecture of switches limits dynamic configuration according to application-specific requirements.
- b. Switch configuration required according to the installed operating system (OS).
- c. Centralized control.
- d. All of the above**

**Correct Answer: d**



---

**Detailed Solution:** see fig. in lec 33, slide 7

---

**QUESTION 14:**

The basic SDN concept involves:

- a. Separate control logic from hardware switches
- b. Define the control logic in a centralized manner
- c. Control the entire network including individual switches
- d. All of the above**

**Correct Answer: d**

**Detailed Solution:** see lec 33, slide 13.

---

**QUESTION 15:**

Control message from a switch to controller in SDN is termed as:

- a. Flow-rule message
- b. PACKET-IN message**
- c. Control message
- d. Data message

**Correct Answer: b**

**Detailed Solution:** see lec 33, slide 20

---

\*\*\*\*\*END\*\*\*\*\*



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## Introduction to Internet of Things

### Assignment- Week 8

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

---

#### **QUESTION 1:**

From the below given options, which is not a characteristic of cloud computing?

- a. Low management effort
- b. Low level generalization of computation**
- c. On-demand network access
- d. Configurable computing resources

**Correct Answer: b**

**Detailed Solution:**

---

#### **QUESTION 2:**

GreenCloud Simulator is designed based on

- a. NS-3
- b. NS-2**
- c. CloudSim
- d. DCSim

**Correct Answer: b**

**Detailed Solution:**

---

#### **QUESTION 3:**

Which of the following is an example of Software-as-a-Service (SaaS) in cloud computing?

- a. Google App**
- b. Windows Azure
- c. GoGrid
- d. Rackspace Cloud Servers



---

**Correct Answer: a**

**Detailed Solution:**

See lecture on Cloud Computing Fundamentals, Slide no 17

---

**QUESTION 4:**

In which of the following cloud deployment model is owned by an organization and offers the cloud service?

- a. Private Cloud
- b. Public Cloud**
- c. Hybrid Cloud
- d. None of the above

**Correct Answer: b**

**Detailed Solution:**

See lecture on Cloud Computing Fundamentals, Slide nos 19-22

---

**QUESTION 5:**

If the devices are highly mobile in IoT, one of the disadvantage of using SDN can be:

- a. High control overhead**
- b. Low data rate
- c. All of the above
- d. None of the above

**Correct Answer: a**

**Detailed Solution:**

---

**QUESTION 6:**

Which of the following is/are components of cloud computing?

- a. Infrastructure
- b. Services



- c. Platform
- d. **All the above**

**Correct Answer: d**

**Detailed Solution:**

See lecture on Cloud Computing Fundamentals, Slide no. 13

---

**QUESTION 7:**

Mice flows have:

- a. Large duration
- b. **Short duration**
- c. Moderately large duration
- d. None of the above

**Correct Answer: b**

**Detailed Solution:**

---

**QUESTION 8:**

Which of the following statement is true?

- a. **SaaS enables the remote access of software via Internet where web-browser acts as a thin-client**
- b. SaaS provides the access of non-commercial software only
- c. SaaS is only service model, which does not support pay-as-use model
- d. SaaS provides a multi-tenant application delivery in a many-to-one model

**Correct Answer: a**

**Detailed Solution:**

See lecture on Cloud Computing-Service Model, Slide nos. 20-21

---

**QUESTION 9:**

Which of the following service model of cloud computing is an **application middleware** offered as a service to developers?





- a. IaaS
- b. PaaS**
- c. SaaS
- d. None of the above

**Correct Answer: b**

**Detailed Solution:**

See lecture on Cloud Computing-Service Model, Slide no. 15

---

**QUESTION 10:**

Which of the following is not used for Database-as-a-service?

- a. ClearDB
- b. DatabaseHome**
- c. Database.com
- d. MicrosoftAzure/SQLDatabase

**Correct Answer: b**

**Detailed Solution:**

See lecture on Cloud Computing – Service Management and Security, Slide no. 12

---

**QUESTION 11:**

Which of the following is an example for building a cloud infrastructure

- a. EUCALYPTUS**
- b. FireFox
- c. SDN
- d. CloudExpress

**Correct Answer: a**

**Detailed Solution:**

[https://en.wikipedia.org/wiki/Eucalyptus\\_\(software\)](https://en.wikipedia.org/wiki/Eucalyptus_(software))

---

**QUESTION 12:**



Access control layers in cloud include

- a. Cloud access
- b. Server access
- c. Service access
- d. **All the above**

**Correct Answer: d**

**Detailed Solution:**

See lecture on Cloud Computing – Service Management and Security, Slide no. 21

---

**QUESTION 13:**

The top most layer of CloudSim architecture

- a. enable modeling and simulation
- b. manages the clock
- c. provide cloud environment
- d. **presents different machines and application specifications**

**Correct Answer: d**

**Detailed Solution:**

See lecture on Cloud Computing – Case Studies, Slide no. 8

---

**QUESTION 14:**

CloudAnalyst is based on

- a. OpenStack
- b. CloudStack
- c. **CloudSim**
- d. EUCALYPTUS

**Correct Answer: c**

**Detailed Solution:**



---

See lecture on Cloud Computing – Case Studies, Slide no. 12

---

**QUESTION 15:**

GreenCloud is

- a. **capable of monitoring the energy consumption of servers**
- b. capable of providing the SaaS only
- c. a commercial cloud platform
- d. oldest cloud platform

**Correct Answer: a**

**Detailed Solution:**

See lecture on Cloud Computing – Case Studies, Slide no. 15

---

\*\*\*\*\*END\*\*\*\*\*



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## Introduction to Internet of Things

### Assignment- Week 9

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

---

#### **QUESTION 1:**

We normally take snapshot in Openstack, \_\_\_\_\_.

- a. **after the installation of an instance**
- b. before the installation of an instance
- c. it is an inbuilt image in Openstack
- d. it should be downloaded from the internet

**Correct Answer: a**

#### **Detailed Solution:**

As a snapshot saves a system state, similar to data backup, it can be taken only after the installation of an instance.

---

#### **QUESTION 2:**

Sensor-Cloud deals with

- a. Sensor-for-Service
- b. **Sensor-as-a-Service**
- c. Sensor-in-Cloud
- d. None of these

**Correct Answer: b**

#### **Detailed Solution:**

evident from lecture 42, title slide (slide-1)

---

#### **QUESTION 3:**



---

In OpenStack, NOVA is

- a. image service
- b. object service
- c. **compute service**
- d. used for providing the networking service

**Correct Answer: c**

**Detailed Solution:**

See lecture on Cloud Computing – Practical, Slide no 6

---

**QUESTION 4:**

In OpenStack, the duration of service usage is monitored by

- a. Cinder
- b. **Ceilometer**
- c. Heat
- d. Neutron

**Correct Answer: b**

**Detailed Solution:**

See lecture on Cloud Computing – Practical, Slide no 8

---

**QUESTION 5:**

The concept of Sensor-Cloud is

- a. based on simple storing the sensor data into cloud
- b. based on the accumulating the sensor data from one application region and transmit these data to another region only
- c. based on virtualization of sensor
- d. None of the above options is correct

**Correct Answer: c**

**Detailed Solution:**



---

See lecture on Sensor-Cloud Part I, Slide no. 10

---

**QUESTION 6:**

Among the following actors in sensor-cloud architecture, who is responsible for managing the infrastructure and deals with price?

- a. End-users
- b. Sensor owners
- c. **SCSP**
- d. None of the above
- e.

**Correct Answer: c**

**Detailed Solution:**

**See the research paper on sensor-cloud-**

S. Misra; S. Chatterjee; M. S. Obaidat, "On Theoretical Modeling of Sensor Cloud: A Paradigm Shift From Wireless Sensor Network," in IEEE Systems Journal, vol.PP, no.99, pp.1-10

---

**QUESTION 7:**

In sensor-cloud, who takes decision whether the data should be provided directly to the end user or not?

- a. **Internal Cache**
- b. External Cache
- c. End-user
- d. Sensor-owner

**Correct Answer: a**

**Detailed Solution:**

See lecture on Sensor-cloud Part II, Slide no. 13

---

**QUESTION 8:**

What is a good approach to reduce latency?

- a. **Analyzing data close to the data source**



- b. Analyzing data in cloud
- c. Increasing number of nodes
- d. Decreasing number of nodes

**Correct Answer: a**

**Detailed Solution:**

see lec-44, slide 14

---

**QUESTION 9:**

Which of the following company coined the term “Fog Computing” ?

- a. Apple
- b. Amazon
- c. **CISCO**
- d. Microsoft

**Correct Answer: c**

**Detailed Solution:**

See lecture on Fog Computing Part I, Slide no. 2

---

**QUESTION 10:**

Resource usage information in service accounting is obtained in the form of

- a. Data set
- b. Tables
- c. Records
- d. Any form

**Correct Answer: c**

**Detailed Solution:**

recall from week 8, lec 39, slide 5

---



---

### **QUESTION 11:**

For which of the following reasons the fog computing should be used?

- a. For storing a huge amount of data in the fog device
- b. While handling time sensitive data, a millisecond can make a huge difference**
- c. In the absence of network connectivity with users' device, the fog device can provide the data
- d. The data rate in a fog device is always very high.

**Correct Answer: b**

**Detailed Solution:**

See lecture on Fog Computing Part I, Slide no. 2

---

### **QUESTION 12:**

Which of the following is/are the advantages of Fog computing?

- a. Reduce the latency during decision making
- b. Data can be stored in local server
- c. Fog nodes can be programmed according to the customer's requirements
- d. All the above**

**Correct Answer: d**

**Detailed Solution:**

See lecture on Fog Computing Part II, Slide no. 14

---

### **QUESTION 13:**

To uninstall the OpenStack, which of the following files is required to be run?

- a. Unistack.sh
- b. unstack.sh**
- c. remstack.sh
- d. uninstallstack.sh





---

**Correct Answer: b**

**Detailed Solution:**

See lecture on Cloud Computing – Practical, Slide no 10

---

**QUESTION 14:**

Why data volume is a problem in IoT-based cloud computing?

- a. because data are encrypted and hard to analyze
- b. because data coming from IoT devices are always in raw format and difficult to store
- c. because the density of IoT devices is increasing each day and in turn the volume of generated data is growing very fast
- d. because IoT device density is decreasing each day and in turn the volume of generated data are decreasing very fast

**Correct Answer: c**

**Detailed Solution:**

see lec-44, slides 4,5,6

---

**QUESTION 15:**

Blue Pill attack on hypervisor is a threat example of:

- a. Host security at PaaS level
- b. Host security at SaaS level
- c. **Host security at IaaS level**
- d. None of the above

**Correct Answer: c**

**Detailed Solution:**

recall from week-8, lec-39, slide 16

---

\*\*\*\*\*END\*\*\*\*\*



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## Introduction to Internet of Things

### Assignment- Week 10

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

---

#### **QUESTION 1:**

Smart Agriculture incorporates:

- a. Alert generation in case of above-threshold pollutants in the air.
- b. Scheduling harvesting and arranging proper transfer of harvests to warehouses or markets.**
- c. Both the above
- d. None of these

**Correct Answer: b**

**Detailed Solution:** see lec 46, slide 14

---

#### **QUESTION 2:**

Heterogeneity for IoT in smart cities stands for:

- a. Integration of varying hardware platforms and specifications.
- b. Integration of different radio specifications.
- c. Integration of various software platforms.
- d. All of these**

**Correct Answer: d**

**Detailed Solution:** see lec 46, slide 16

---

#### **QUESTION 3:**

Which company first coined fog computing?

- a. IBM
- b. HP
- c. Dell
- d. CISCO**

**Correct Answer: d**

**Detailed Solution:** recall from previous week's lec 44, slide 2

---

#### **QUESTION 4:**

Which of these connects cable-based networks with wireless networks for increased sharing of media, control and access and can share network media resources?

- a. UPnP
- b. DLNA**
- c. LAN
- d. WiFi



**Correct Answer: b**

**Detailed Solution:** see lec 48, slide 11

---

**QUESTION 5:**

The need for smart cities arose due to:

- a. Rapidly growing urban population
- b. Fast depleting natural resources
- c. Changes in environment and climate
- d. **All of the above**

**Correct Answer: d**

**Detailed Solution:** see lecture 46, slide 2

---

**QUESTION 6:**

The following goals are currently associated with smart vehicles:

- a. Assistance to drivers during bad weather or low-visibility.
- b. Detection of bad driving patterns or driving under the influence of substances.
- c. Auto alert generation during crashes.
- d. **All of the above**

**Correct Answer: d**

**Detailed Solution:** see lecture 46, slide 12.

---

**QUESTION 7:**

Which of the following is a social and legal challenge of IoT in smart cities?

- a. Services based on user provided information may be subject to local or international laws.
- b. Individual and informed consent required for using humans as data sources.
- c. **Both a and b**
- d. None of these

**Correct Answer: c**

**Detailed Solution:** see lecture 46, slide 18.

---

**QUESTION 8:**

In multi-sensor data fusion, which is true:

- a. Combines information from multiple sensor sources.
- b. Enhances the ability of decision making systems to include a multitude of variables prior to arriving at a decision.
- c. Inferences drawn from multiple sensor type data is qualitatively superior to single sensor type data.
- d. **All of the above**

**Correct Answer: d**



---

**Detailed Solution:** see lec-47, slide 3.

---

**QUESTION 9:**

Amount of energy harvested depends upon:

- a. Sensitivity of the harvesting circuit
- b. Distance between the device and source
- c. Environment
- d. **All of the above**

**Correct Answer: d**

**Detailed Solution:** See lec47, slide 18.

---

**QUESTION 10:**

X-10 protocol is characterized by:

- a. **Low-speed and low data rate.**
- b. Low speed and high data rate
- c. High speed and low data rate
- d. High speed and high data rate

**Correct Answer: a**

**Detailed Solution:** See lec48, slide 15.

---

**QUESTION 11:**

UPnP stands for:

- a. United Protocol and Plugs
- b. Universal Protocol and Plugs
- c. **Universal Plug and Play**
- d. United Plug and Play

**Correct Answer: c**

**Detailed Solution:** See lec48, slide 10.

---

**QUESTION 12:**

Konnex (KNX) is:

- a. **an open standard for home and building networks.**
- b. Smart parking system
- c. Smart energy system
- d. None of these

**Correct Answer: a**

**Detailed Solution:** see lec-48, slide 12.

---

**QUESTION 13:**

HAN architectures:



- 
- a. Uses XML for description and web-services for control.
  - b. Follows a Service oriented Architecture (SOA).
  - c. Not tied to any software, language or architecture.
  - d. **All of these**

**Correct Answer: d**

**Detailed Solution:** see lec 48, slide 16

---

**QUESTION 14:**

Which of these can be considered as the skeleton for smart cities?

- a. **Buildings**
- b. Transportation
- c. Banks
- d. Sensors

**Correct Answer: d**

**Detailed Solution:** see lec 46, slide 3.

---

**QUESTION 15:**

DSRC in the context of smart vehicles stands for:

- a. Detailed Short Rate Communication
- b. **Dedicated Short Range Communication**
- c. Data Short Range Communication
- d. Data Spedy Robust Communication

**Correct Answer: b**

**Detailed Solution:** see lec 50, slide 3

---

\*\*\*\*\*END\*\*\*\*\*



## Introduction to Internet of Things

### Assignment- Week 11

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

---

#### **QUESTION 1:**

Self-healing in smart grid can be done using –

- a. Dampening unwanted power oscillations
- b. Avoiding unwanted flows of current through the grid
- c. Rerouting power flows in order to avoid overloading in a transmission line
- d. All of these**

**Correct Answer: d**

**Detailed Solution:** see lec 52, slide 4

---

#### **QUESTION 2:**

Renewable energy is generated from –

- a. Natural resources**
- b. Artificial resources
- c. Nuclear resources
- d. All of the above

**Correct Answer: a**

**Detailed Solution:** see lec 46, slide 16

---

#### **QUESTION 3:**

The V2X paradigm follows:

- a. A distributed architecture**
- b. A centralized architecture
- c. Both a and b
- d. None of these

**Correct Answer: a**

**Detailed Solution:** recall week 10, lec 49, slide 6

---

#### **QUESTION 4:**

What is the key technology to enable V2V and V2R communication?

- a. RFID Communication
- b. Dedicated Short Range Communication**
- c. Long Range Communication
- d. Medium Range Communication

**Correct Answer: b**



---

**Detailed Solution:** see lec 54, slide 5

---

**QUESTION 5:**

What are the risks associated with IIoT?

- a. **Lack of vision and leadership**
- b. Cheap sensors
- c. Confident workforce
- d. Both cheap sensors and confident workforce

**Correct Answer: a**

**Detailed Solution:** see lec 54, slide 21

---

**QUESTION 6:**

RSUs are connected to the Internet by:

- a. Relays
- b. Routers
- c. **Gateways**
- d. None of these

**Correct Answer: c**

**Detailed Solution:** recall week 10, lec 50, slide 14

---

**QUESTION 7:**

IIoT is mainly focused on

- a. Convenience of individuals
- b. **Efficiency, safety and security of operation and individuals.**
- c. Efficiency of the system.
- d. Data security.

**Correct Answer: b**

**Detailed Solution:** see lec 53, slide 15

---

**QUESTION 8:**

Smart grid enables –

- a. **Distributed energy management**
- b. Centralized energy management
- c. Both distributed and centralized energy management
- d. None of these

**Correct Answer: a**

**Detailed Solution:** see lec 51, slide 7. It is only distributed management of energy.

---

**QUESTION 9:**

Traditional electric grids are associated with:



- a. Energy generation is done in centralized power plants
- b. Energy distribution is one directional – from the power plant to the homes or industries.
- c. Uni -directional communication
- d. **All of these**

**Correct Answer: d**

**Detailed Solution:** See lec51, slide 2.

---

**QUESTION 10:**

The smart grid is also termed as:

- a. Electricity with a brain
- b. The energy internet
- c. The electronet
- d. **All of these**

**Correct Answer: d**

**Detailed Solution:** See lec51, slide 3

---

**QUESTION 11:**

Which of the following is not a property associated with power quality of smart grids:

- a. Self -Healing
- b. Frequency Monitoring and Control
- c. Load Forecasting
- d. **Asset Management**

**Correct Answer: d**

**Detailed Solution:** See lec51, slide 8

---

**QUESTION 12:**

Home power generation is a term associated with:

- a. Power generation system at consumers -end
- b. Rooftop solar electric systems
- c. Small wind turbines
- d. **All of these**

**Correct Answer: d**

**Detailed Solution:** See lec51, slide 17

---

**QUESTION 13:**

In smart grids, PMU stands for:

- a. Phase measurement unit
- b. **Phasor measurement unit**
- c. Phase monitoring unit
- d. All of these





**Correct Answer: b**

**Detailed Solution:** see lec 52, slide 3

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**QUESTION 14:**

The typical data rate associated with EnOcean is?

- a. **9600 bps**
- b. 3200 bps
- c. 19200 bps
- d. 1152300 bps

**Correct Answer: a**

**Detailed Solution:** see lec 52, slide 9.

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**QUESTION 15:**

MDMS in the context of smart grids stands for:

- a. **Meter data management unit**
- b. Metric data measurement unit
- c. Metric data monitoring unit
- d. None of these

**Correct Answer: a**

**Detailed Solution:** see lec 52, slide 8

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\*\*\*\*\*END\*\*\*\*\*



## Introduction to Internet of Things

### Assignment- Week 12

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

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#### **QUESTION 1:**

If all the data points are same, statistical dispersion becomes –

- a. 1
- b. 10
- c. 100
- d. None

**Correct Answer: d**

**Detailed Solution:** see lec-56, slide 15. In this case dispersion is 0.

---

#### **QUESTION 2:**

The desirable characteristics of an IoT sensor node are

- a. Energy-efficiency
- b. Distributed sensing
- c. Low-cost
- d. All of the above

**Correct Answer: d**

**Detailed Solution:** all three options - a,b,c are highly desirable in a sensor node, which would allow the node to be deployed for long operational hours spread over large areas in a dense manner.

---

#### **QUESTION 3:**

Long term trend tracking and preventive care in IoT healthcare is possible due to

- a. ability of smart devices to collect data autonomously
- b. ubiquitous connectivity
- c. smart algorithms and analytics
- d. all of the above

**Correct Answer: d**

**Detailed Solution:** refer to lec-58, slide 8. Long term tracking is possible only if devices detect events and collect data smartly, and are always connected so as to provide alerts in cases of emergencies. Smart algorithms enable detection of events.

---

#### **QUESTION 4:**

Activity monitoring in smart homes:

- a. Provides better quality of life
- b. Provides information accurately in a reliable manner



- c. Provides continuous monitoring support.
- d. All of these**

**Correct Answer: d**

**Detailed Solution:** All three options a,b,c are true for smart homes.

---

**QUESTION 5:**

Network-based activity recognition is:

- a. Larger and processing intensive
- b. Low power consumption
- c. Dependent on network connection
- d. All of these**

**Correct Answer: d**

**Detailed Solution:** see lec 59, slide 9

---

**QUESTION 6:**

Which of these is a feature of qualitative data analytics?

- a. Data is not described through numerical values
- b. Described by some sort of descriptive context such as text
- c. Data can be gathered by many methods such as interviews, videos and audio recordings, field notes.
- d. All of these**

**Correct Answer: d**

**Detailed Solution:** see lec 56, slide 2

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**QUESTION 7:**

Which is true?

- a. The statistical model is defined as the mathematical equation that are formulated in the form of relationships between variables.
- b. A statistical model illustrates how a set of random variables is related to another set of random variables.
- c. Both a and b**
- d. None of these

**Correct Answer: c**

**Detailed Solution:** see lec 56, slide 8

---

**QUESTION 8:**

ANOVA is applied

- a. when less than 2 samples/populations are to be compared
- b. when more than 2 samples/populations are to be compared**
- c. Both a and b



d. None of these

**Correct Answer: b**

**Detailed Solution:** see lec 56, slide 11.

---

**QUESTION 9:**

Which of the following can be the reasons for packet drop in IoT network?

- a. Dense foliage
- b. High temperature
- c. Rain
- d. All of the above**

**Correct Answer: d**

**Detailed Solution:** All of these (a,b,c) cause some kind of interference with low power radios such as Bluetooth, Zigbee, etc., which are used in IoT networks.

---

**QUESTION 10:**

The types of ANOVA are:

- a. 2
- b. 3**
- c. 4
- d. 5

**Correct Answer: b**

**Detailed Solution:** See lec 56, slide 13

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**QUESTION 11:**

Wearables can be used for:

- a. Child -care
- b. Elderly -care
- c. Entertainment
- d. All of these**

**Correct Answer: d**

**Detailed Solution:** See lec59, slide 2

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**QUESTION 12:**

Wearable-based activity monitoring can be used for:

- a. Continuous monitoring of activity results in daily observation of human behavior and repetitive patterns in their activities.
- b. Easy integration and fast equipping
- c. Long term monitoring
- d. All of these**



---

**Correct Answer: d**

**Detailed Solution:** See lec59, slide 5

---

**QUESTION 13:**

Which of these is true is case of network based activity monitoring?

- a. Larger and processing intensive methods can be applied
- b. Group based analytics possible
- c. **Both a and b**
- d. None of these

**Correct Answer: c**

**Detailed Solution:** see lec 59, slide 9

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**QUESTION 14:**

What is the network topology of Ambusens WBAN devices?

- a. Bus topology
- b. **Star Topology**
- c. Mesh Topology
- d. None of the above

**Correct Answer: b**

**Detailed Solution:** see figure in lec 58, slide 16

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**QUESTION 15:**

Which of these sensors can be most cheaply used for basic activity monitoring in wearables?

- a. **Accelerometer**
- b. Cameras
- c. LIDARs
- d. LED

**Correct Answer: a**

**Detailed Solution:** cameras and LIDARs are very costly, and cannot be generally used with wearables. LED is not a sensor. The only plausible option is accelerometers (these are actually present in fitness tracker arm-bands.)

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\*\*\*\*\*END\*\*\*\*\*